



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____
Date / /

- DOHC - Double Overhead Camshaft
- PFD - Primary Flight Display
- MFD - Multi-Function Display

Battery:

- ES - 28V
- Batt - 24V
- Alternator - 70 Amp
- Main Batt - 24V

- Tricycle type Fixed landing gear
- Air-oil Type struts
- 6.00x6, Four ply rated wheels.

Positions:

- Landing gear: nose wheel using rudders and brakes 3rdy each side
- Elevator trim: b/w front seats on the floor
- Rudder trim: mounted on the right side of the pedestal below throttle
- Flaps: manually operated, spring loaded to return to the up position, past centre lock. 3 positions: 10, 25, 40°



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date _____/_____/_____

1) • FADEC - Full Authority Digital Engine Control

• CED 125 - Compact Engine Display

• AED 125 - Auxiliary Engine Display

• G1000 - Garmin 1000

* FADEC - It is a computer system that manages and controls all aspects of all A/C engine performance.

2) Warning -

Caution -

✓ 3) Engine Manufacture - Technify Motors GmbH

✓ Engine Model - TAE 125-02-114

Engine Mount - Tabular steel

4) Engine - • Liquid cooled

• In-line four-stroke 4 cylinder turbocharged engine with DOHC (Double overhead camshaft)

• Direct fuel injection and common-rail technology.

• Displacement of 121.5 cubic inches.



5.) Not available service:

- Carburetor and carburetor pre-heating.
- Ignition magnetos and spark plugs.
- Mixture control and priming system.

6.) Engine warning:

- If the main battery and alternator fail, the engine will only operate on A-FADEC for max. 30 mins on FADEC backup battery power.

7.) Propeller: (Wooden composite)

- ✓ • manufacture - MT Propeller Entwicklung GmbH
- ✓ • model - MTV-6-A/190-69
- ✓ • No. of Blades - 3
- Diameter - 1.90 m (74.8 in)
- ✓ • Type - variable-pitch propeller (constant speed)

✓ 8.) Fuel - JET-A-1/Jet-A, Diesel

9.) Engine oil - Aeroshell Oil Diesel Ultra.



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date ____/____/____

10.) Gearbox oil - Shell Spirax S6 G XME 75W-80

11.) Coolant - Water / Radiator Protection at a ratio of 50:50

▲ warning - The engine must not be started under any circumstances if any fluid level is too low.

• Freezing point of the coolant is -36°C .

12.) Engine operating Limits:

✓ • Tlo and max. continuous Power - 114 KW (155hp)

✓ • Tlo and max. continuous RPM - 2300 rpm

✓ • Max. recommended cruise - 85%

• Economic cruise $\rightarrow$ 70% load.

▲ Starting the engine with external power is not allowed.

• Change of the original A/C is certified up to an altitude ~~16400~~ 16400 ft (5000m).

• Engine load $\rightarrow$ Available % of max. engine power.



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date / /

13) Oil temperature:

- min engine starting temp - -32°C
- min engine operating limit temp - 50°C
- Max. operating limit temp - 140°C

14) Coolant temperature:

- min engine starting temp - -32°C
- min. operating limit temp. - 60°C
- Max. operating limit temp. - 105°C
- Freezing point of coolant - -36°C

15) Gearbox temperature:

- min. operating limit temp - -30°C
- max. operating limit temp - 120°C

16) Oil pressure:

- min oil Px - $1.2 \text{ bar} (14.5 \text{ PSI})$
- min oil Px (at T/O power) - $2.3 \text{ bar} (33.4 \text{ PSI})$
- min oil Px (in flight) - $2.3 \text{ bar} (33.4 \text{ PSI})$
- max. oil Px - $6.0 \text{ bar} (87 \text{ PSI})$
- max oil Px (cold start $< 20 \text{ sec}$) - 6.5 bar
- max. oil consumption - 0.1 l/hr



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date / /

17) Weight Limits :

- Max. Ramp weight - 1160 Kg - 2558 lbs
- Max. T/O weight - 1157 Kg
- Max. Landing weight - 1157 Kg

18) Max. Fuel Quantities : (Fuel tank switched every 30 min)

PA-28-181 : [• While starting - Left, • Drain - Each tank fuel strainer]

- 2 standard tanks - each 94.6 L
- Total capacity - 189.3 L (50 USG)
- Total usable fuel - 181.7 L (48 USG)
- Total unusable fuel - 7.6 L (2 USG)
- Reserve - 45 min. Fuel warning below - 10 liters (2.6 USG)

19) Engine load - Available percentage of max. Engine Power,

- Max. Turning radius from pivot point to wing tip - 30 ft
- Nose gear steering $\rightarrow 30^\circ$ & later A/c 20°
- Wing loading - 15 lbs/sq.ft
- Power loading - 14.2 lbs/hp



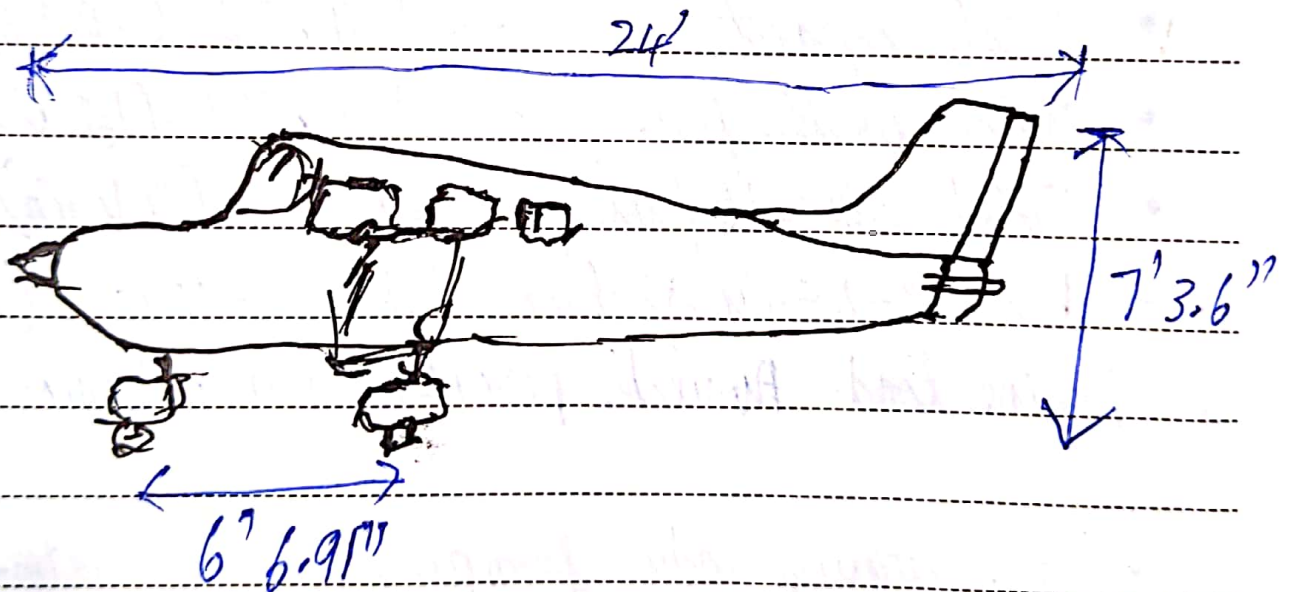
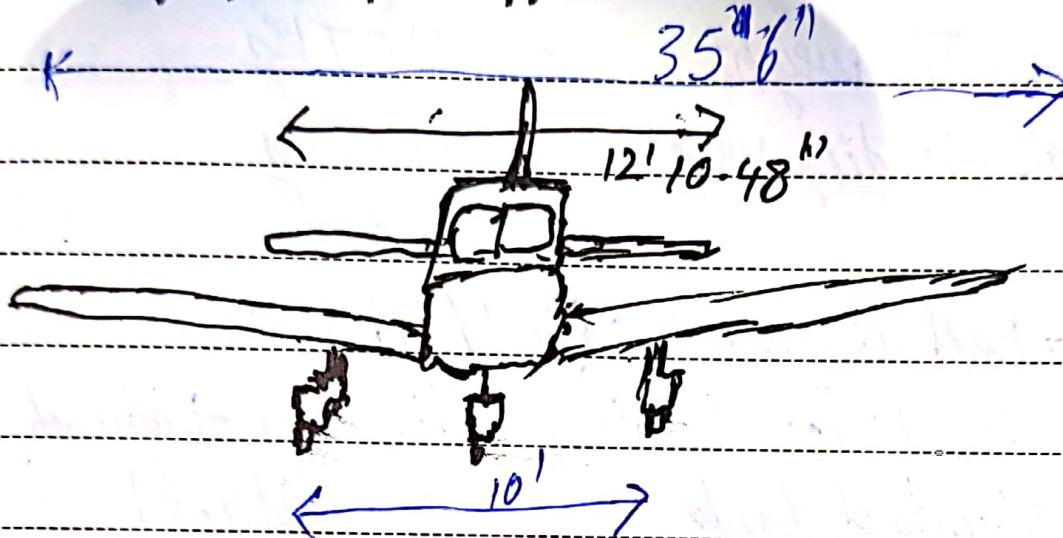
Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date / /

-20- General measurements:

- wing span (Tip to Tip) - $35'6''$





Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date / /

21.) Terms :

- CAS - Indicated speed of an A/C corrected for position and instrument error.
- GS - speed of an airplane relative to the ground.
- IAS - speed of an A/C shown on ASI corrected for instrument error.
- TAS - Airspeed of an A/C relative to undisturbed air which is the CAS corrected for altitude, temperature and compressibility.
- V_0 - max. operating maneuvering speed is the max. speed at which application of full available aerodynamic control will not overstress the airplane.
- V_{FE} - Max. Flap extended speed is the highest speed permissible with wing flaps in a prescribed extended position.

22.) ASI markings :

- | | | |
|---------------------------|-------------|-------------------------|
| • Red Line (Never Exceed) | - 154 | Final Approach speed |
| • Yellow Band | - 125 - 154 | at Flaps 40° - 63 Kts |
| • Green Band | - 50 - 125 | Best rate of climb |
| • White band (Flap Down) | - 45 - 104 | with flaps 10° - 65 Kts |



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date / /

23) V_r - rotation speed - 55 Kts

V_x - Best angle of climb - 64 Kts

V_y - Best rate of climb - 76 Kts

V_{FE} - max. flaps extend speed - 102 Kts

V_A - Manoeuvring speed - 113 Kts

V_{NO} - normal operating speed - 125 Kts

V_{NE} - Never exceed speed - 154 Kts / 155 Kts

V_{SO} - ^{Dirty} ~~Clean~~ configuration (landing) - 45 Kts

V_{S1} - Stall speed in clean configuration - 50 Kts

V - 75 Kts

Approach speed for 10° Flap - 70 Kts

Approach speed for 40° Flap - 66 Kts

Glide speed without flap - 73 Kts

Glide speed with flaps - 65 Kts

• V_A - Manoeuvring speed - 113 KIAS at 2550 lbs

• Best glide speed - 78 KIAS

• Main battery voltage - 24V



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date _____/_____/_____

24.) Max. certified Alt - 16400ft

25.) Fuel capacity -

26.)

- If the propeller speed control fails, climbs are performed at - 65 KIAS & 100% Load
- During Engine failure after T/O airspeed with flaps retracted should be - 73 KIAS
- In an Engine failure, the propeller will continue to turn as long as the airspeed is above - 65 ~~48~~ KIAS

27.) 2500 lbs (1157 Kg) ÷

- Lift off speed - 55 KIAS
- After T/O, speed at 15m (50ft) - 60 KIAS
- V_y - 76 KIAS

28.) TYER Pressure → Nose Gear - 18 PSI
→ Main Gear - 24 PSI

29.) ELT testing freq - 121.5 MHz



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date / /

• During ELT testing, the pulses should be restricted to 3 pulses.

• Height loss during stall varies from 100 to 350 ft
302 stall warning horn is activated - 5-10 Kts above stall speed.

31.) • Max. allowed baggage in baggage compartment is - 200 lbs

• Baggage area is - 24 Cu in.

32.) Warm up procedure is - 890 rpm for 2 min, then 1400 rpm till OT 50 deg C and CT 60 deg C

33.) • Starter motor can be operated for max of - 10 sec per attempt

• After 6 attempt - 1/2 hr (30 min) given to cool starter motor

34.) Fuel tanks must be switched every - 30 mins

35.) Alternator RED Range - 760 Amps.



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date / /

36.) Gear box temp - Green - $< 115^{\circ}\text{C}$

Amber - $115-120^{\circ}\text{C}$

Red - $> 120^{\circ}\text{C}$

37.) Tachometer - Green - $0-2300\text{ RPM}$

Red - $> 2300\text{ RPM}$

38.) Oil pressure - Green - $2.3-5.1\text{ bar}$

Amber - $1.1-2.3\text{ bar}$

Red - $0-1.1\text{ bar} \text{ \& } > 6.5\text{ bar}$

39.) Oil temp - Green - $50-129^{\circ}\text{C}$

Amber -

Red -

40.) Coolant temp - Green - $60-100^{\circ}\text{C}$

Amber -

Red -



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date / /

41) Fuel Temp - green - 0-65deg C

42) FADEC warning in the G1000 will display - ECU A Fail
ECU B Fail

43) Engine coolant level low will show on the
G1000 as - 200k 2VL - ~~low~~ caution.

44) When Force B is activate, the display on the
G1000 shown - ECU B Active.

45) Tire px - Front wheel - 18 PSI
Rear wheel - 24 PSI

46) Wing loading - 15 lbs/sq ft
Power loading - 14.2 lbs/hp

47) Flaps setting - (3) $\begin{matrix} 10^\circ \\ 25^\circ \\ 40^\circ \end{matrix}$

Strut exposure - Nose wheel - $3.25 \pm .25$ in

Rear wheel - $4.5 \pm .25$ in



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date / /

48.) wing loading - Gross weight per surface area of wing
Power loading - Total gross weight per total power

PFD - Instrument - CDI

- VOR - $1\text{dot} = 5^\circ$, full deflection $= 10^\circ$ or more
- ILS - $1\text{dot} = 1.25^\circ$, full deflection $= 2.5^\circ$ or more

Bearing - Towards the stn.

Radial - Away from stn.

T/O roll -

- $\downarrow 10\%$ for each 9 kts H.W
- $\uparrow 10\%$ for each 2 kts T.W

• Flaps are manually operated & spring loaded to return to the $\rightarrow$ up position

• Short field T/O flaps config $\rightarrow 25^\circ$

• Normal T/O: Flaps 25° / 79 kts climb